

Random Variable: A random variable is a function that associates a real number with each element in the sample space.

In other words, a random variable is a function $X: S \rightarrow R$, where S is the sample space of the random experiment under consideration.

Note: Normally a capital letter, say X , is used to denote a random variable and its corresponding small letter, x in this case, for one of its values.

Example: Consider the random experiment of tossing a coin three times and observing the result (a Head or a Tail) for each toss. Let X denote the total number of heads obtained in the three tosses of the coin.

- (i) Construct a table that shows the values of the random variable X for each possible outcome of the random experiment.
- (ii) Identify the event $\{X \leq 1\}$ in words.

Let Y denote the difference between the number of heads obtained and the number of tails obtained.

- (iii) Construct a table showing the value of Y for each possible outcome.
- (iv) Identify the event $\{Y = 0\}$ in words.

Discrete Random Variable: If a sample space contains a finite number of possibilities or an unending sequence with as many elements as there are whole numbers (countable), it is called a discrete sample space. A random variable is called a discrete random variable if its set of possible outcomes is countable.

Example: (i) Two balls are drawn in succession without replacement from an urn containing 4 red balls and 3 black balls. The possible outcomes and the values x of the random variable X , where X is the number of red balls, are

Sample Space	x
RR	2
RB	1
BR	1
BB	0

Example (ii) Suppose that our experiment consists of tossing 3 fair coins. If we let X denote the number of heads appearing, then X is a random variable taking on one of the values 0, 1, 2, 3 with respective probabilities

$$S = \{(H, H, H), (H, H, T), (H, T, H), (T, H, H), \\ (T, T, H), (T, H, T), (H, T, T), (T, T, T)\}$$

$$P\{X = 0\} = P\{(T, T, T)\} = 1/8$$

$$P\{X = 1\} = P\{(T, T, H), (T, H, T), (H, T, T)\} = 3/8$$

$$P\{X = 2\} = P\{(T, H, H), (H, T, H), (H, H, T)\} = 3/8$$

$$P\{X = 3\} = P\{(H, H, H)\} = 1/8$$

Discrete Probability Distributions: A discrete random variable assumes each of its values with a certain probability.

In the case of tossing a coin three times, the variable X , representing the number of heads, assumes the value 2 with probability $3/8$, since 3 of the 8 equally likely sample points result in two heads and one tail. The possible values x of X and their probabilities are

X	0	1	2	3
$P(X=x)$	$1/8$	$3/8$	$3/8$	$1/8$

Probability Mass Function: Let X be a one dimensional discrete random variable which takes the values $x_1, x_2, x_3, \dots$. Then $P(X = x_i) = P(x_i)$ satisfies the following conditions

1. $P(x_i) \geq 0$

2. $\sum_{i=1}^{\infty} P(x_i) = 1$

Cumulative Distribution Function of Discrete Random Variable X:

The distribution function of a discrete random variable X defined in $(-\infty, \infty)$ is given by

$$F(x) = P(X \leq x) = \sum_{t \leq x} f(t), \quad -\infty < x < \infty$$

Properties of the Distribution function

1. $P(a < X \leq b) = F(b) - F(a)$
2. $P(a \leq X \leq b) = P(X = a) + F(b) - F(a)$
3. $P(a < X < b) = F(b) - F(a) - P(X = b)$
4. $P(a \leq X < b) = F(b) - F(a) - P(X = b) + P(X = a)$

Example(i) A random variable X has the following probability function

Value of X , x_i	0	1	2	3	4	5	6	7	8
Probability $P(x)$	a	$3a$	$5a$	$7a$	$9a$	$11a$	$13a$	$15a$	$17a$

1. Determine the value of ' a '.
2. Find $P(X < 3)$, $P(X \geq 3)$. $P(0 < X < 5)$.
3. Find the distribution function of X .

1. Since $\sum_{i=1}^{\infty} P(x_i) = 1$

$$a + 3a + 5a + 7a + 9a + 11a + 13a + 15a + 17a = 1$$

$$a = \frac{1}{81}$$

2. $P(X < 3) = P(0) + P(1) + P(2) = a + 3a + 5a = 9a = 1/9$

$$P(X \geq 3) = 1 - P(X < 3) = 8/9$$

$$P(0 < X < 5) = P(1) + P(2) + P(3) + P(4)$$

$$= 3a + 5a + 7a + 9a = 24a = 24/81$$

3.

x	0	1	2	3	4	5	6	7	8
P(x)	a	3a	5a	7a	9a	11a	13a	15a	17a
F(x)	a	4a	9a	16a	25a	36a	49a	64a	81a

Alternate method for sub-division 2, using the cumulative distribution function $F(x)$.

$$P(X < 3) = P(X \leq 2) = F(2) = 9a = 1/9$$

$$P(X \geq 3) = 1 - P(X < 3) = 1 - (1/9) = 8/9$$

$$\begin{aligned} P(0 < X < 5) &= F(5) - F(0) - P(X = 5) \\ &= 36a - a - 11a \\ &= 24a \\ &= 24/81 \end{aligned}$$

Example (ii) A random variable X has the following probability function

Value of X , x_i	0	1	2	3	4	5	6	7
Probability $P(x)$	0	a	$2a$	$2a$	$3a$	a^2	$2a^2$	$7a^2 + a$

1. Determine the value of ' a '.
2. Find $P(1.5 < X < 4.5 / X > 2)$.
3. Find the smallest value of λ for which $P(X \leq \lambda) > 1/2$.

$$1. \text{ Since } \sum_{i=1}^{\infty} P(x_i) = 1$$

$$10a^2 + 9a = 1$$

$a = 1/10$ or -1 . As $a = -1$ is meaningless, $a = 1/10$

$$\begin{aligned} 2. \quad P(1.5 < X < 4.5 / X > 2) &= \frac{P[(1.5 < X < 4.5) \cap (X > 2)]}{P(X > 2)} \\ &= \frac{P(X = 3) + P(X = 4)}{1 - [P(X = 0) + P(X = 1) + P(X = 2)]} = \frac{5}{7} \end{aligned}$$

$$3. \quad P(X \leq 0) = 0; P(X \leq 1) = 0.1; P(X \leq 2) = 0.3; \\ P(X \leq 3) = 0.5 \text{ and } P(X \leq 4) = 0.8$$

$\therefore \lambda = 4$ for which $P(X \leq \lambda) > 1/2$.

Continuous Random Variable: If a random variable takes on all values within a certain interval, then the random variable is called Continuous random variable.

E.g., The height, age and weight of individuals, the amount of rainfall on a rainy day.

Probability Density Function: If X is a continuous random variable then $f(x)$ is called the probability density function of X provided $f(x)$ satisfies the following conditions;

1. $f(x) \geq 0, \forall x$

2. $\int_{-\infty}^{\infty} f(x)dx = 1$

Cumulative Probability Distribution of Continuous Random Variable X : The cumulative distribution function $F(x)$ of a continuous random variable X with density function $f(x)$ is

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt, \quad -\infty < x < \infty$$

Results:

(a) 1. $P(a \leq x \leq b) = \int_a^b f(x)dx = F(b) - F(a)$

2. When X is continuous r.v.

$$P(X = a) = P(a \leq X \leq a) = \int_a^a f(x)dx = 0$$

$$\therefore P(a < X \leq b) = P(a \leq X < b) = P(a < X < b) = P(a \leq X < b).$$

(b) If $F(x)$ is the distribution function of one dimensional random variables, then

1. $0 \leq F(x) \leq 1$

2. If $x < y$, then $F(x) \leq F(y)$

3. $F(-\infty) = 0$, $F(\infty) = 1$.

4. If X is discrete r.v. taking values $x_1, x_2, x_3, \dots$ where $x_1 < x_2 < x_3 < \dots$ then $P(X = x_i) = F(x_i) - F(x_{i-1})$.

5. If X is continuous r.v., then $\frac{dF(x)}{dx} = f(x)$

Example If the density function of a continuous r.v. X is given by

$$f(x) = \begin{cases} ax, & 0 \leq x \leq 1 \\ a, & 1 \leq x \leq 2 \\ 3a - ax & 2 \leq x \leq 3 \\ 0, & \text{elsewhere} \end{cases}$$

1. Find the value of a
2. Find the cumulative distribution function of X
3. Find $P(1.5 < X \leq 3)$
4. Find $P(X > 1.5)$

Solution: 1. Since $f(x)$ is a p.d.f.

$$\int_{-\infty}^{\infty} f(x) dx = 1 \rightarrow \int_0^3 f(x) dx = \int_0^1 ax dx + \int_1^2 a dx + \int_2^3 (3a - ax) dx = 1$$

$$a = 1/2.$$

2. $F(x) = P(X \leq x) = 0, \quad x < 0$

$$F(x) = \int_0^x ax dx = \left[\frac{ax^2}{2} \right]_0^x = \frac{ax^2}{2} = \frac{x^2}{4}, \quad 0 \leq x \leq 1$$

$$\begin{aligned} F(x) &= \int_0^1 ax dx + \int_1^x a dx \\ &= \left[\frac{ax^2}{2} \right]_0^1 + [ax]_1^x \\ &= \left[\frac{a}{2} \right] + [ax - a] = ax - \frac{a}{2} = \frac{x}{2} - \frac{1}{4}, \quad 1 \leq x \leq 2 \end{aligned}$$

$$\begin{aligned}
F(x) &= \int_0^1 a x dx + \int_1^2 a dx + \int_2^3 (3a - ax) dx \\
&= \left[\frac{ax^2}{2} \right]_0^1 + [ax]_1^2 + \left[3ax - \frac{ax^2}{2} \right]_2^x \\
&= \left[\frac{a}{2} \right] + a + \left[\left(3ax - \frac{ax^2}{2} \right) - (6a - 2a) \right] \\
&= 3ax - \frac{ax^2}{2} - \frac{5a}{2} \\
&= \frac{1}{4} (6x - x^2 - 5), \quad 2 \leq x \leq 3
\end{aligned}$$

$$F(x) = P(X \leq x) = 1, \quad x > 3$$

2. Cumulative Distribution function

$$F(x) = P(X \leq x) = \begin{cases} 0, & x < 0 \\ \frac{x^2}{4}, & 0 \leq x \leq 1 \\ \frac{1}{4} + \frac{x-1}{2}, & 1 \leq x \leq 2 \\ \frac{-5}{4} + \frac{6x-x^2}{4}, & 2 \leq x \leq 3 \\ 1, & x > 3 \end{cases}$$

$$3. \quad F(1.5) = \frac{1}{4} + \frac{x-1}{2} = \frac{1}{4} + \frac{1.5-1}{2} = \frac{1}{2}, \quad 1 \leq x \leq 2$$

$$F(3) = \frac{-5}{4} + \frac{6x - x^2}{4} = -\frac{5}{4} + \frac{6(3) - (3)^2}{4} = 1, \quad 2 \leq x \leq 3$$

$$\therefore P(1.5 < x < 3) = F(3) - F(1.5) = 1 - \frac{1}{2} = \frac{1}{2}$$

$$4. \quad \begin{aligned} P(X > 1.5) &= \int_{1.5}^3 f(x) dx \\ &= \int_{1.5}^2 \frac{1}{2} dx + \int_2^3 \left(\frac{3}{2} - \frac{x}{2} \right) dx \\ &= \frac{1}{2} \end{aligned}$$

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THANK YOU